Confira os códigos-fonte e os materiais da disciplina:

https://github.com/unipds-engenharia-de-ia-aplicada

Abaixo são as referências e links complementares às aulas, dividido por módulos e submódulos:

Disciplina 01 - Fundamentos de IA e LLMs para Programadores

1.1 - Módulo - Introdução

Jogo do pacman:

- https://storage.googleapis.com/tfjs-examples/webcam-transfer-learning/dist/index.html

Wear Presentations:

- https://vimeo.com/177959241
- https://codelabs.developers.google.com/magicwand#0
- https://www.kaggle.com/models/google/mobilenet-v1/
- https://www.datacamp.com/tutorial/machine-deep-learning
- https://www.datacamp.com/tutorial/machine-deep-learning

1.1.1 Teachable Machine - Treinando um modelo a partir de um dataset

- https://teachablemachine.withgoogle.com/
- https://kaggle.com/
- https://www.kaggle.com/competitions/titanic
- https://www.kaggle.com/competitions/titanic/data
- https://microsoft.github.io/hack-workshop-lobe/
- https://medium.com/@raphael.semeteys/a-guide-to-node-js-notebooks-461344cf4feb
- https://lobe-ai-website-clone.vercel.app/
- https://www.kaggle.com/datasets/gpiosenka/70-dog-breedsimage-data-set

1.2 - Módulo - Machine Learning, Deep Learning e IA - a base de tudo

- https://www.notablecap.com/blog/the-anatomy-of-a-neural-network
- https://research.google/blog/sensorlm-learning-the-language-of-wearable-sensors/
- https://www.notablecap.com/blog/the-anatomy-of-a-neural-network
- https://research.google/blog/sensorlm-learning-the-language-of-wearable-sensors/

1.3 - Módulo - Deep Learning - Sistemas de Recomendação na prática

1.3.1

Projeto em exemplo00

- https://poloclub.github.io/cnn-explainer/
- https://poloclub.github.io/ganlab/
- https://github.com/jessevig/bertviz?tab=readme-ov-file
- https://teachablemachine.withgoogle.com/
- https://netron.app/
- https://www.tensorflow.org/tensorboard
- https://ai.google.dev/edge/litert/web
- https://ai.google.dev/edge/litert/conversion/tensorflow/build/ondevice_training

Simulated Spotify Listening Experiences for Reinforcement Learning with TensorFlow and TF-Agents

- https://blog.tensorflow.org/2023/10/simulated-spotify-listening-experiences-reinforcement-learning-tensorflow-tf-agents.html

1.3.2 Como Construir um Sistema de Recomendação Inteligente com Redes Neurais no Navegador

Projeto em exemplo-01-ecommerce-recomendations-z

- https://github.com/recommenders-team/recommenders

- https://www.databricks.com/blog/guide-to-building-online-recommendation-system

1.4 - Módulo - Web Machine Learning - Como Vencer Qualquer Jogo

Projeto em exemplo-02-vencendo-qualquer-jogo

- https://github.com/ErickWendel/DuckHunt-JS/tree/ml-self-play-template
- https://github.com/ErickWendel/DuckHunt-JS/tree/ml-self-play
- https://www.linkedin.com/in/hugozanini/
- https://blog.tensorflow.org/2022/05/real-time-sku-detection-in-browser.html
- https://medium.com/data-science/training-a-custom-yolov7-in-pytorch-and-running-it-directly-in-the-browser-with-tensorflow-js-96a5ecd7a530
- https://docs.ultralytics.com/datasets/detect/coco/#coco-pretrained-models
- https://github.com/ultralytics/yolov5
- https://docs.ultralytics.com/integrations/tfjs/
- https://github.com/MattSurabian/DuckHunt-JS?tab=readme-ov-file
- https://pixijs.com/
- https://caniuse.com/?search=using
- https://dev.to/zacharylee/explicit-resource-management-in-js-the-using-keyword-d9f

5 - Módulo - Inteligência artificial na Web

5.1 - Algoritmos Genéticos

- https://rednuht.org/genetic_cars_2/
- https://youtu.be/P7XHzqZjXQs?si=ZhcSK9Ism2zhe4fX&t=1453
- https://youtu.be/r8KWciNmEGw?si=Qq_fquK5P1lh9AGN&t=862
- https://github.com/prouhard/tfjs-mountaincar
- https://medium.com/@pierrerouhard/reinforcement-learning-in-the-browser-an-introduction-to-tensorflow-js-9a02b143c099
- https://storage.googleapis.com/tfjs-examples/cart-pole/dist/index.html
- https://github.com/tensorflow/tfjs-examples/tree/master/cart-pole
- https://github.com/jinglescode/reinforcement-learning-tic-tac-toe?tab=readme-ov-file
- https://medium.com/data-science/reinforcement-learning-implement-tictactoe-189582bea542

- https://github.com/rfeinman/tictactoe-reinforcement-learning
- https://www.freecodecamp.org/news/how-to-build-an-adaptive-tic-tac-toe-ai-with-reinforcement-learning-in-javascript/
- https://github.com/MJeremy2017/reinforcement-learning-implementation/tree/master
- https://github.com/mayur9210/tic-tac-toe-ai/tree/main
- https://rednuht.org/genetic_cars_2/
- https://medium.com/codesphere-cloud/teaching-cars-to-drive-with-neuroevolution-tensorflow-and-500-lines-of-javascript-57888956322e
- https://github.com/jonatan5524/SnakeAI?tab=readme-ov-file
- https://www.youtube.com/watch?v=cdUNkwXx-I4

5.2 - Como funcionam LLMs: transformers, embeddings, attention

- https://www.youtube.com/watch?v=ynNDmp7hBdo&list=PLqFwRPueWb5cUZEMIFlDuODaOYeSTp31V
- https://youtu.be/qFoFKLI3O8w?si=ZtMgIcbu0WtW0-7T
- https://www.emailvendorselection.com/chatgpt-statistics/
- https://technext24.com/2023/08/14/chatgpt-costs-700000-daily-openai/
- https://www.reuters.com/commentary/breakingviews/openai-takes-its-cut-expanding-ai-boom-2025-10-06/
- https://www.youtube.com/watch?v=NKnZYvZA7w4

5.3 - Web AI - Como funciona inteligência artificial na Web com Web AI

Projeto em exemplo-03-webai01

Projeto em exemplo-04-webai02-temperature-and-topK

- https://x.com/jason_mayes?lang=en
- https://youtu.be/IC256KyITLw?si=XhuOSVmBEZ6LOjVw&t=62
- https://huggingface.co/webml-community
- https://huggingface.co/spaces/webml-community/kokoro-web
- https://huggingface.co/spaces/webml-community/deepseek-r1-webgpu
- https://developer.chrome.com/docs/ai/built-in?hl=pt-br
- https://developer.chrome.com/docs/ai/built-in-apis
- https://chrome.dev/web-ai-demos/
- https://googlechromeai2025.devpost.com/

- https://googlechromeai2025.devpost.com/project-gallery
- https://x.com/nic_o_martin
- https://github.com/nico-martin?tab=repositories
- https://github.com/nico-martin/ask-my-pdf
- https://github.com/nico-martin/benz-gpt
- https://github.com/nbonamy/witsy
- https://ai.google.dev/gemini-api/docs/prompting-strategies

5.4 - Web AI - AI multimodal direto em navegadores

Projeto em exemplo-05-webai03-multimodal/

- https://github.com/webmachinelearning/webmcp/blob/main/README.md
- https://www.youtube.com/watch?v=ka2Zzw_qors&list=PLqFwRPueWb5dAcgxDGg7egpIhGRpcgiLN
- https://github.com/ErickWendel/Trimly

6 - Módulo - Prompt Engineering na prática

6.1 - Como escrever prompts que realmente geram boas respostas

- https://youtu.be/ysPbXH0LpIE
- https://platform.openai.com/tokenizer
- https://platform.openai.com/docs/guides/prompt-engineering
- https://www.anthropic.com/engineering/effective-context-engineering-for-ai-agents
- https://platform.claude.com/docs/en/build-with-claude/prompt-engineering/claude-4-best-practices
- https://docs.lovable.dev/prompting/prompting-debugging

6.2 - Padrão TOON e JSON para prompts

- https://github.com/toon-format/toon?tab=readme-ov-file
- https://mpgone.com/json-prompt-guide/
- https://apidog.com/pt/blog/json-format-prompts-pt/
- https://medium.com/data-science-in-your-pocket/toon-bye-bye-json-for-llms-91e4fe521b14

- https://github.com/toon-format/toon
- https://toontools.vercel.app/playground
- https://www.linkedin.com/posts/webai_toon-json-mcp-activity-7395932385500069888-vgRC?utm_source=share&utm_medium=member_desktop&rcm=ACoAABS1ckYBm0_ZMP8TT2e4opa3bFMxTRxyJ3w
- https://www.linkedin.com/posts/pawel-huryn_how-to-format-data-in-prompts-for-llms-and-activity-7397245245299761152-VGHQ

7 - Módulo - Ferramentas de IA para acelerar sua vida como Dev

7.1 - História e visão geral do Cursor, Comparando VSCode, Windsurf e Cursor

- https://www.latent.space/p/cursor
- https://news.crunchbase.com/venture/cursor-financing-ai-coding-automation/
- https://news.crunchbase.com/venture/cursor-financing-ai-coding-automation/
- https://sacra.com/c/cursor/
- https://techcrunch.com/2024/08/29/github-copilot-competitor-codeium-raises-150m-at-a-1-25b-valuation/
- https://www.reuters.com/business/openai-agrees-buy-windsurf-about-3-billion-bloomberg-news-reports-2025-05-06/
- https://github.com/features/copilot/plans
- https://code.visualstudio.com/docs/copilot/customization/custom-agents
- https://fortune.com/2025/07/23/ai-coding-tool-replit-wiped-database-called-it-a-catastrophic-failure/

7.2 - O que são agentes de IA e como eles tomam decisões em etapas

- https://developer.microsoft.com/blog/spec-driven-development-spec-kit
- https://www.marcohaber.dev/blog/git-worktrees

8 - Módulo - MCPs e automação para devs

8.1 - O que são MCPs (Model Context Protocol) e como conectam IA com APIs/serviço

Projeto em erickwendel-contributions-mcp/

- https://www.anthropic.com/
- https://www.anthropic.com/news/model-context-protocol
- https://modelcontextprotocol.io/
- https://cookbook.openai.com/examples/reasoning_function_calls
- https://github.com/resend/mcp-send-email/tree/main
- https://erickwendel.com.br/
- https://tml-api.herokuapp.com/graphiql

8.2 - Usando IA para Gerar testes automatizados

Projeto em exemplo-06-playwright-testes/

- https://github.com/microsoft/playwright-mcp
- https://playwright.dev/docs/test-agents#agent-definitions
- https://www.linkedin.com/feed/update/urn:li:activity:7381279589840433152/

8.3 - Usando IA para navegação

Projeto em exemplo-07-playwright-navegacao/

8.4 - Usando IA para consultar documentações atualizadas

Projeto em exemplo-08-context7/

- https://github.com/upstash/context7#installation
- https://github.com/ChromeDevTools/chrome-devtools-mcp
- https://context7.com/
- https://github.com/upstash/context7
- https://context7.com/websites/nodejs_api
- https://nextjs.org/
- https://www.better-auth.com/

8.5 - Usando IA para colher dados de telemetria de apps

Projeto em exemplo-09-grafana-mcp/

9 - Módulo - Modelos open-source vs. proprietários

9.1 - Vantagens e desvantagens de modelos abertos e fechados (e sem censura)

- https://github.com/torch2424/wasmboy?tab=readme-ov-file#wasmboy
- https://github.com/eugeneyan/open-llms
- https://ollama.com/library/gpt-oss
- https://ollama.com/library/llama3.1
- https://ollama.com/library/gemma3
- https://ollama.com/search?q=uncensored
- https://openai.com/open-models/
- https://fortune.com/2023/07/18/meta-llama-2-ai-open-source-700-million-mau/

9.2 - Como rodar modelos localmente com Ollama

Projeto em exemplo-10-ollama/

- https://ollama.com/search
- https://docs.vllm.ai/en/latest/
- https://ollama.com/library/gpt-oss
- https://openai.com/index/introducing-gpt-oss/
- https://huggingface.co/mlc-ai/Llama-3-8B-Instruct-q4f32_1-MLC
- https://llm.mlc.ai/docs/get_started/introduction.html
- https://cdn.openai.com/pdf/419b6906-9da6-406c-a19d-1bb078ac7637/oai_gpt-oss_model_card.pdf
- https://ollama.com/blog/vision-models
- https://docs.ollama.com/integrations/vscode
- https://www.jan.ai/docs/desktop/llama-cpp-server

9.3 - Como usar OpenRouter para orquestrar vários modelos (e parar de empilhar “licenças”)

Projeto em exemplo-11-openrouter/

- https://openrouter.ai/
- https://openrouter.ai/models?max_price=0
- https://github.com/marketplace?type=models

10 - Módulo - RAG, embeddings e busca semântica

10.1 - O que é RAG (Retrieval-Augmented Generation) e por que ele é tão importante

- https://arxiv.org/pdf/2005.11401

10.2 - Embeddings e Vector Databases na prática

Projeto em exemplo-12-embeddings-neo4j/

- https://neo4j.com/
- https://docs.langchain.com/oss/javascript/integrations/text_embedding/transformers

10.3 - Criando o primeiro RAG com JavaScript e Neo4j

Projeto em exemplo-13-embeddings-neo4j-rag/

- https://docs.langchain.com/oss/javascript/langchain/overview

---

Problemas no Windows

- Falha ao instalar o @tensorflow/tfjs-node